Arduino code:

```
// Define the Arduino pins
int ECGPin = A0;
int LOminusPin = 10;
int LOplusPin = 11;

// Set the sampling frequency
int sampleRate = 250;

// Time between samples in microseconds
unsigned long sampleTime = 1000000 / sampleRate;
unsigned long nextSampleTime = 0;

void setup() {

  // Start serial communication
  Serial.begin(115200);

  // Set lead-off pins as inputs
  pinMode(LOminusPin, INPUT);
  pinMode(LOplusPin, INPUT);

  // Set the first sampling time
  nextSampleTime = micros();
```

```
}

void loop() {

  // Check whether it is time to take another sample
  if ((long)(micros() - nextSampleTime) >= 0) {

    // Schedule the next sample
    nextSampleTime += sampleTime;

    // Read the ECG signal
    int ECGvalue = analogRead(ECGPin);

    // Read the lead-off pins
    int LOminus = digitalRead(LOminusPin);
    int LOplus = digitalRead(LOplusPin);

    // Send the measurements to MATLAB
    Serial.print(ECGvalue);
    Serial.print(",");
    Serial.print(LOminus);
    Serial.print(",");
    Serial.println(LOplus);

  }
```

```
}
```